How well does the Surgery Risk Model predict?

Out-of-time test on 19,986 patients eligible on 2025-07-01, a quarter the model never saw.

The bottom line

0.73

AUC: ranks a surgery patient above a non-surgery patient 73% of the time

2.16%

predicted risk, against 2.16% observed: calibrated on a new quarter

42%

of surgeries caught by contacting the top 2,500 (12.5% of patients)

What this means

- The model's risk estimates can be taken at face value: across risk groups, predicted and observed surgery rates match.
- Contacting the top 2,500 reaches 3.3× as many future surgeries as a random list of the same size.
- Surgery is rare (2.2% a year), so most listed patients won't have surgery: the list concentrates risk, it doesn't guarantee it.

How the test works

- Train: 60,594 patient snapshots, quarterly from 2024-07 to 2025-04.
- Test: the 2025-07 snapshot. 19,986 eligible patients, 431 of whom had surgery while covered within 12 months.
- The model never sees the test quarter, so this is how it performs on the next one Meridian will score.
- Eligible: covered, back-pain history, no surgery yet, no surgical consult in the last 90 days.
- Precision and recall use a fixed list of the top 2,500, about the size of Meridian's past waves. How many to actually contact is a separate decision (the 'How far down the list' deck).
- 95% intervals come from resampling test patients 500 times.

Metrics on the test quarter

Metric	Value	95% interval	What it means
AUC	0.733	0.704 – 0.761	Chance a surgery patient ranks above a non-surgery one
Mean predicted risk	2.16%	2.13% – 2.20%	Average of the model's predictions
Observed surgery rate	2.16%	1.95% – 2.36%	What actually happened
Precision	7.2%	6.3% – 8.2%	Surgery rate among listed patients (vs 2.2% overall)
Recall (surgeries caught on the list)	41.8%	37.4% – 46.0%	Share of all surgeries in the top 2,500
F1	0.123	0.108 – 0.139	Balance of precision and recall
Accuracy	87.1%	86.9% – 87.3%	Misleading at a 2.2% rate: listing nobody scores 97.8%

Who the top 2,500 catches, and who it misses

	Had surgery	No surgery
On the list	180	2,320
Not on the list	251	17,235

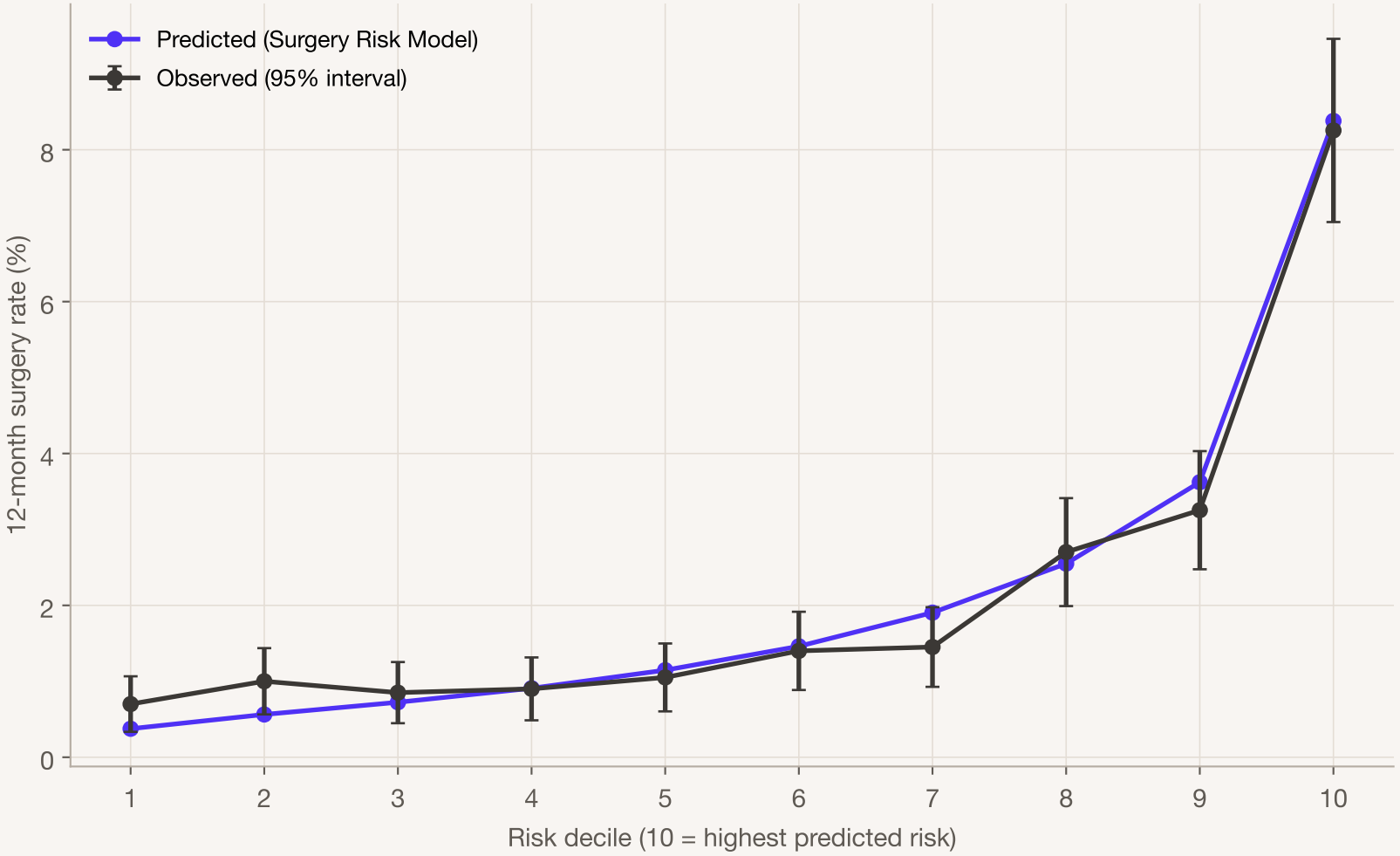
- 180 of the 431 surgeries are on the list:

  recall 42%.
- 180 of 2,500 listed patients had surgery:

  precision 7.2%, 3.3× the base rate.
- 251 surgeries came from patients ranked below 2,500.

  Some surgeries happen to patients who look low-risk.

Predicted risk matches what happened

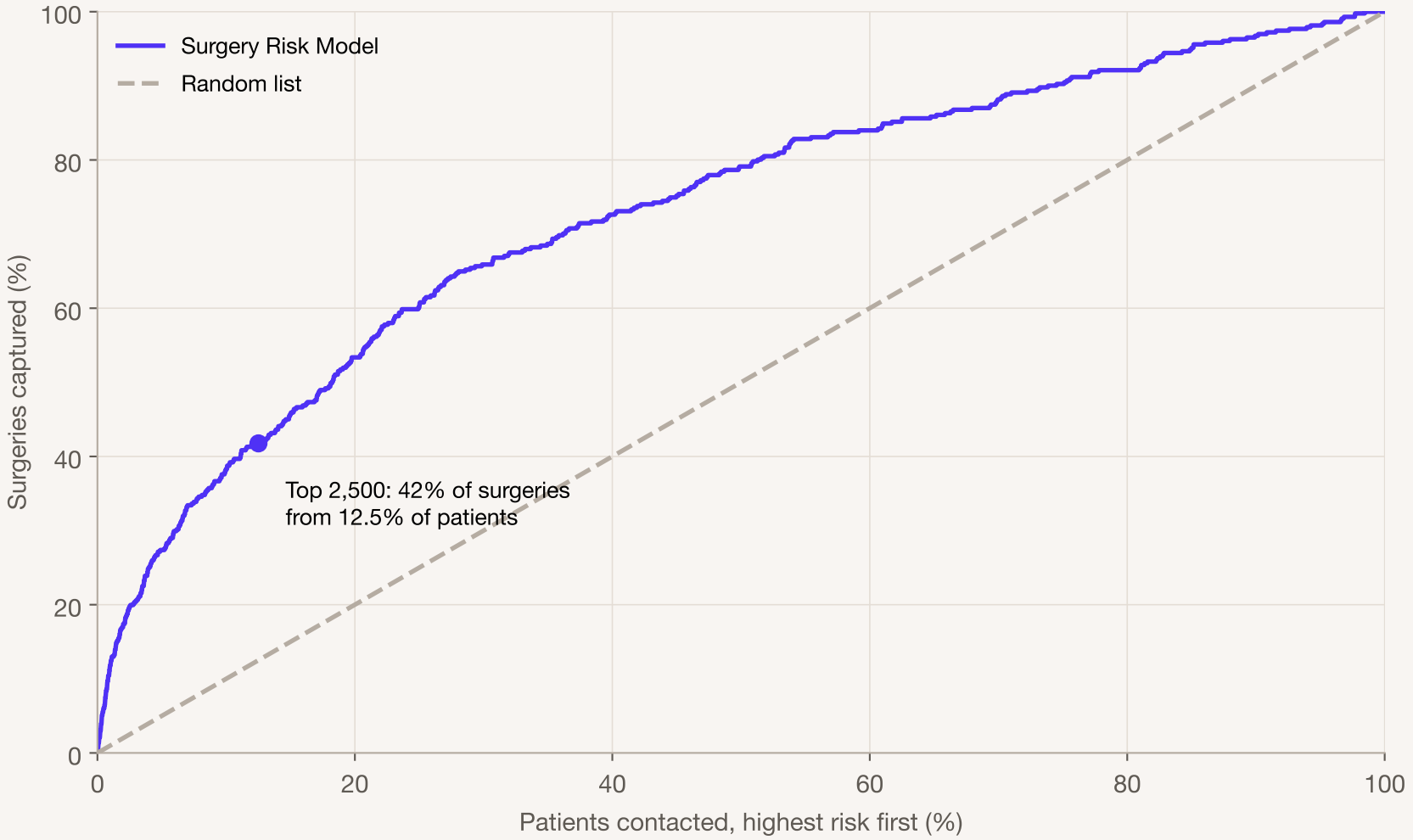


Each point is a tenth of the test patients, sorted by risk.

Top decile: 8.4% predicted, 8.3% observed.

Bottom deciles run slightly low (0.4% vs 0.7%), which doesn't affect who makes the list.

A 2,500-patient list catches 42% of surgeries



Contact patients in risk order and count how many of the year's surgeries you've reached.

A random list of 2,500 catches 12.5%. This one catches 42%.

The curve flattens: past the top third, extra contacts find few extra surgeries.

Five real patients from the test quarter

Situation	Profile	Predicted risk	Rank	On list	Outcome
Top of the list, had surgery	57M, BMI not recorded, prior consult, recent imaging	35.3%	3	Yes	Surgery
High risk, no surgery	78M, BMI not recorded, prior consult, recent imaging	24.3%	21	Yes	No surgery
On the cut-off line (rank 2,500)	75F, BMI not recorded, prior consult	4.0%	2,500	Yes	No surgery
Typical patient	76M, BMI 32	1.3%	9,994	No	No surgery
Missed: low risk, had surgery	77M, BMI not recorded	0.3%	19,677	No	Surgery

A predicted risk is a probability, not a verdict: most high-risk patients don't have surgery within a year, and some low-risk ones do.

Confident enough to rank Meridian's list

- Calibrated on a quarter it never saw: 2.16% predicted vs 2.16% observed.
- Ranks well: AUC 0.73, and the top 2,500 catch 42% of surgeries.
- Limits: half of next quarter's patients are 24+ months past their first visit, beyond the training range, so their risks may run slightly high. Re-check calibration every quarter.